Event Boundaries Directionally Influence Item-Level Recognition Memory: A Replication Using the Mnemonic Similarity Task

BRSM 2026 — Report 1

Team: *Behavioral Segments*

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Code: https://github.com/ashkehcum/MST-Mid-Report.git

1 Introduction

Our daily experiences unfold as a continuous stream of information, yet we remember them as discrete chunks—or *events*—separated by *event boundaries* (Zacks et al., 2007). Event boundaries arise when an observer perceives a meaningful change in context, goals, or perceptual flow. Research consistently shows that these boundaries shape memory organisation: temporal-order memory is impaired across boundaries relative to within-event comparisons (DuBrow & Davachi, 2013; Ezzyat & Davachi, 2014; Clewett et al., 2019), and items near boundaries often receive privileged processing (Swallow et al., 2009).

While the influence of event boundaries on *temporal* memory is well established, far less is known about how they affect the *fidelity* of item-level recognition memory. Morse et al. (2023) addressed this gap by combining an incidental encoding paradigm with a **Mnemonic Similarity Task** (MST; Stark et al., 2019; Kirwan & Stark, 2007). The MST probes two complementary aspects of recognition:

1. **Target recognition**—the ability to correctly identify previously studied items as "old," indexed by the Recognition Memory Index REC = $P(\text{“old”} \mid \text{Target}) - P(\text{“old”} \mid \text{Foil})$.
2. **Lure discrimination**—the ability to reject perceptually similar but non-identical *lures*, thought to depend on hippocampal pattern separation (Yassa & Stark, 2011; Bakker et al., 2008). This is indexed by the Lure Discrimination Index LDI = $P(\text{“similar”} \mid \text{Lure}) - P(\text{“similar”} \mid \text{Foil})$.

Morse et al. (2023) reported two directional effects. First, basic target recognition was *selectively impaired* for items encoded immediately *after* a boundary ("post-boundary"), while pre-boundary and mid-event items did not differ. Second, lure discrimination was *selectively enhanced* for items encoded immediately *before* a boundary ("pre-boundary"). Critically, these effects emerged only when boundaries were defined by a conjunction of a perceptual shift (change in item category) *and* a

task shift (change in encoding question), not by either factor alone.

The present study is a replication of Morse et al. (2023) with the following modifications: (a) the encoding question "natural or manmade?" was replaced by "will it fit in a shoebox?"; (b) lure-bin similarity labels (1–5) were incorporated; (c) events contained 7 items each (40 events = 280 encoding trials); and (d) the test phase comprised 60 targets, 60 lures, and 30 foils.

Three experimental groups were tested:

- **Item Only** ($n = 56$): Objects and scenes with a single fixed encoding question ("indoor or outdoor?") and no task switch at boundaries.
- **Both** ($n = 49$): Objects and scenes with a task switch at every boundary ("shoebox?" for objects, "indoor/outdoor?" for scenes)—analogous to Experiment 1 of Morse et al. (2023).
- **Task Only** ($n = 53$): Objects only with a task switch at every boundary (alternating "shoebox?" and "indoor/outdoor?" applied to objects)—analogous to Experiment 3 of Morse et al. (2023).

The key prediction, following Morse et al. (2023), is that the *Both* group (perceptual + task shift) should show the strongest directional boundary effects, while the *Item Only* group (no task switch) should show no effect, and the *Task Only* group (task switch without perceptual category change) should show intermediate or null effects.

2 Methods

2.1 Participants

A total of 158 participants were included across three groups: Item Only ($n = 56$), Both ($n = 49$), and Task Only ($n = 53$). No exclusion criteria were applied.

2.2 Materials and Procedure

Encoding phase. Participants viewed 280 images divided into 40 events of 7 items each. Within each event the position of each item was coded relative to event boundaries: **post-boundary** (position 0, immediately after a boundary), **mid-event** (positions 1–5), and **pre-boundary** (position 6, immediately before the next boundary). Groups differed in the type of boundary experienced (see Introduction). All tasks were run in PsychoPy via a web browser.

Test phase. After encoding, participants completed a surprise recognition test with 150 trials: 60 targets (previously studied items, ending in `a.jpg`), 60 lures (similar but not identical, ending in `b.jpg`), and 30 foils (completely novel items from a `Foils` folder). Participants responded "old," "similar," or "new" to each image. Targets and lures were balanced across pre-boundary, mid-event, and post-boundary positions (20 each). Lures were drawn from five similarity bins (1 = most similar to 5 = least similar), following the standard MST stimulus sets (Stark et al., 2019).

2.3 Data Processing

Encoding RT. If a participant responded while the image was on screen, RT = `key_resp_9.rt`. If the response occurred after the image disappeared (3 s display time), RT = 3 + `key_resp_8.rt`.

Encoding accuracy. For scenes, accuracy was recomputed against a ground-truth mapping (`scenes_mappin` rather than relying on the CSV-stored `encoding_task_accuracy` field, which contained known errors. For objects ("fits in a shoebox?"), no ground-truth mapping was available, so the CSV accuracy was retained as a reference.

Test-phase classification. Each test trial was classified as a target, lure, or foil based on the image filename. Responses (`o`, `s`, `n`) and response times were extracted from `key_resp_3` columns. Boundary position (`pre/mid/post/none`) was read from the `position_of_stimuli` column. For lures, the similarity bin (1–5) was determined by matching the image number to the `Set6 bins.txt` (objects) or `SetScC bins.txt` (scenes) lookup tables.

2.4 Dependent Measures

- **REC** (Recognition Memory Index): REC = $P(\text{"old"} \mid \text{Target}) - P(\text{"old"} \mid \text{Foil})$. A bias-corrected hit rate; higher values indicate better target recognition.
- **LDI** (Lure Discrimination Index): LDI = $P(\text{"similar"} \mid \text{Lure}) - P(\text{"similar"} \mid \text{Foil})$. Corrects for response bias toward "similar"; higher values indicate better mnemonic discrimination (Stark et al., 2019).
- **Encoding RT**: Mean response time during the encoding phase, computed per boundary position.

All three metrics were computed both overall and separately for each boundary position (pre, mid, post). LDI was additionally computed per lure-similarity bin.

2.5 Statistical Analysis

Within-group comparisons. To test whether boundary position affected memory or RT within each group, *paired-samples* t-tests were performed on all pairwise position contrasts (post vs. pre, post vs. mid, pre vs. mid). The key prediction is that only *one* position differs from the other two: post-boundary impairment for REC (post $<$ mid $\approx$ pre), pre-boundary enhancement for LDI (pre $>$ mid $\approx$ post), and post-boundary slowing for RT (post $>$ mid $\approx$ pre). If only post vs. pre differs but post vs. mid and pre vs. mid are non-significant, the effect is bidirectional rather than selective.

Between-group comparisons. To test whether boundary effects differed across groups, *independent-samples* t-tests compared per-participant difference scores (e.g., $\text{REC}_{\text{post}} - \text{REC}_{\text{pre}}$) between all group pairs. Effect sizes are reported as Cohen's d. All tests are two-tailed and uncorrected for multiple comparisons in this initial analysis; future work will apply appropriate corrections (e.g., Bonferroni or FDR).

3 Results

3.1 Encoding Phase: Response Time

A robust post-boundary RT increase was observed exclusively in the groups that experienced a task switch (Figure 1). In the **Both** group, post-boundary RT ($M = 1.92\,\text{s}$) was significantly higher than both mid-event RT ($M = 1.70\,\text{s}$; $t(48) = 10.01$, $p < .001$, $d = 1.43$) and pre-boundary RT ($M = 1.68\,\text{s}$; $t(48) = 9.32$, $p < .001$, $d = 1.33$), while pre and mid did not differ ($p = .27$). The **Task Only** group showed a virtually identical pattern: post ($M = 2.40\,\text{s}$) was significantly slower than mid ($M = 2.16\,\text{s}$; $t(52) = 9.39$, $p < .001$, $d = 1.29$) and pre ($M = 2.13\,\text{s}$; $t(52) = 8.51$, $p < .001$, $d = 1.17$), with no pre–mid difference ($p = .16$). The **Item Only** group showed no significant RT differences across positions (all $p > .08$).

The per-group mean RT boundary effect (post − pre) was: Both ($M_\Delta = +0.24\,\text{s}$), Task Only ($M_\Delta = +0.27\,\text{s}$), and Item Only ($M_\Delta = -0.05\,\text{s}$). Between groups, this post-boundary RT cost was significantly larger in Both vs. Item Only ($\Delta M = +0.24\,\text{s}$, $t = 8.18$, $p < .001$, $d = 1.60$) and Task Only vs. Item Only ($\Delta M = +0.25\,\text{s}$, $t = 8.06$, $p < .001$, $d = 1.54$), with no difference between Both and Task Only ($p = .64$).

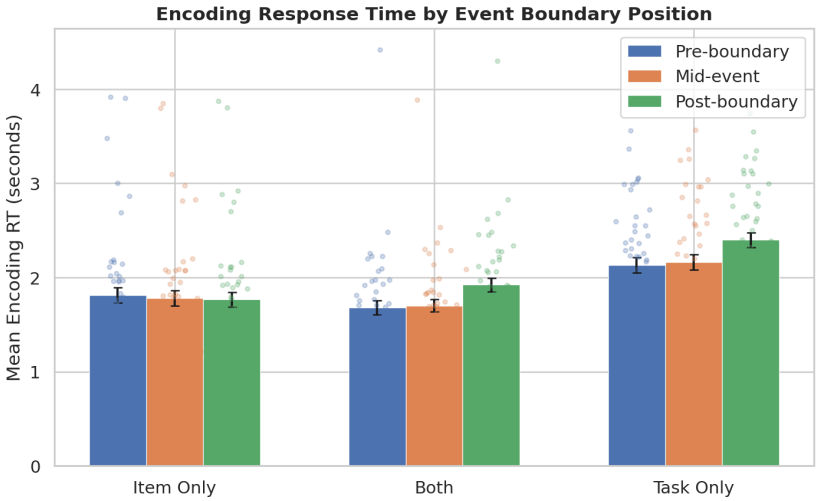


Figure 1: Mean encoding RT by boundary position and group. Error bars represent ±1 SEM. Individual participant data are overlaid.

The within-event RT time course (Figure 2) confirms that elevated RT is sharply localised to position 0 (post-boundary) in the Both and Task Only groups, returning to baseline by position 1.

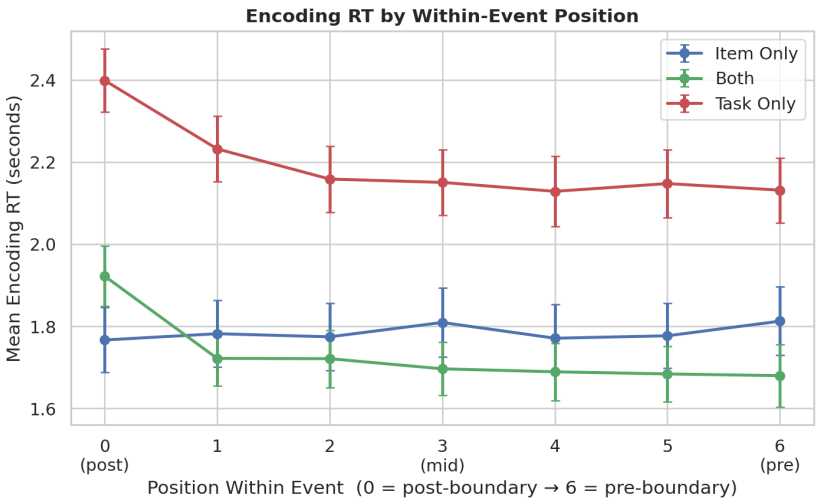


Figure 2: Encoding RT by within-event position (0–6), averaged across all 40 events and all participants per group. Position 0 is post-boundary; position 6 is pre-boundary.

3.2 Recognition Memory (REC)

Target recognition followed the predicted directional pattern in the boundary-active groups (Figure 3). In the **Both** group, REC was significantly lower for post-boundary items ($M = 0.56$) than for both pre-boundary ($M = 0.61$; $t(48) = -2.54$, $p = .015$, $d = -0.36$) and mid-event items ($M = 0.63$; $t(48) = -3.34$, $p = .002$, $d = -0.48$), while pre and mid did not differ ($p = .28$). This confirms a *selective post-boundary impairment*: recognition is reduced specifically for the first item after a boundary, not for pre-boundary items.

The **Task Only** group showed a similar but slightly weaker pattern: post ($M = 0.56$) was significantly lower than pre ($M = 0.62$; $t(52) = -2.69$, $p = .010$, $d = -0.37$), and marginally lower than mid ($M = 0.60$; $t(52) = -1.80$, $p = .078$, $d = -0.25$). Pre and mid did not differ ($p = .17$).

The **Item Only** group showed no significant differences across any position pair (all $p > .49$), confirming that without a task switch, no post-boundary recognition impairment occurs.

The per-group mean REC boundary effect (post − pre) quantifies the magnitude of this impairment: Task Only ($M_\Delta = -0.061$), Both ($M_\Delta = -0.051$), and Item Only ($M_\Delta = +0.007$). Between groups, this boundary effect was significantly larger (more negative) in Both vs. Item Only ($t = -2.11$, $p = .037$, $d = -0.41$) and Task Only vs. Item Only ($t = -2.33$, $p = .022$, $d = -0.45$), with no difference between Both and Task Only ($p = .74$). That is, both task-switch groups showed a meaningful post-boundary recognition cost of roughly 5–6 percentage points, whereas Item Only showed essentially no cost. Rank-ordering the groups by impairment magnitude: **Task Only** showed the largest REC boundary effect ($M_\Delta = -0.061$), followed by **Both** ($M_\Delta = -0.051$), while **Item Only** showed no effect ($M_\Delta = +0.007$). Notably, Task Only and Both did not differ from each other ($p = .74$), but both differed significantly from Item Only, indicating that the presence of a *task switch*—regardless of whether a perceptual category change also occurred—is the critical driver of post-boundary recognition impairment.

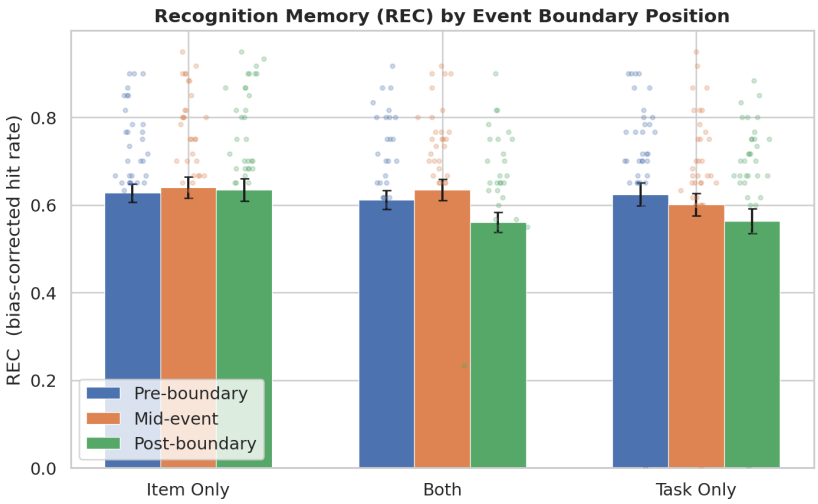


Figure 3: REC (bias-corrected target recognition) by boundary position and group. Error bars represent ±1 SEM with individual data overlaid.

3.3 Lure Discrimination (LDI)

Unlike in Morse et al. (2023), no significant pre-boundary LDI enhancement was found in any group (Figure 4). All within-group pairwise comparisons were non-significant: Item Only (all $p > .41$), Both (all $p > .15$), and Task Only (all $p > .33$). The per-group mean LDI boundary effect (post − pre) was: Both ($M_\Delta = -0.003$), Task Only ($M_\Delta = -0.002$), and Item Only ($M_\Delta = +0.000$). Between-group comparisons of these position-based LDI effects were likewise non-significant (all $p > .09$).

Overall LDI differed between groups: Task Only ($M = 0.38$) was significantly higher than Both ($M = 0.27$; $t = -3.04$, $p = .003$, $d = -0.60$), likely reflecting the objects-only stimulus set in Task Only (objects may be more discriminable than scenes).

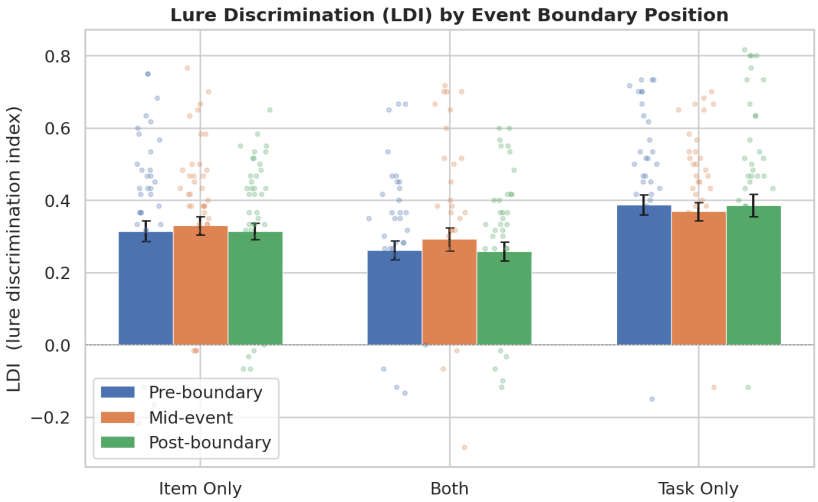


Figure 4: LDI (lure discrimination) by boundary position and group. Error bars represent ±1 SEM with individual data overlaid.

3.4 LDI by Lure Similarity Bin

As expected from previous MST work (Stark et al., 2019), LDI increased monotonically from bin 1 (most similar lures, hardest to discriminate) to bin 5 (least similar lures) across all groups (Figure 5).

This validates the lure difficulty manipulation and confirms that participants were performing the task meaningfully.

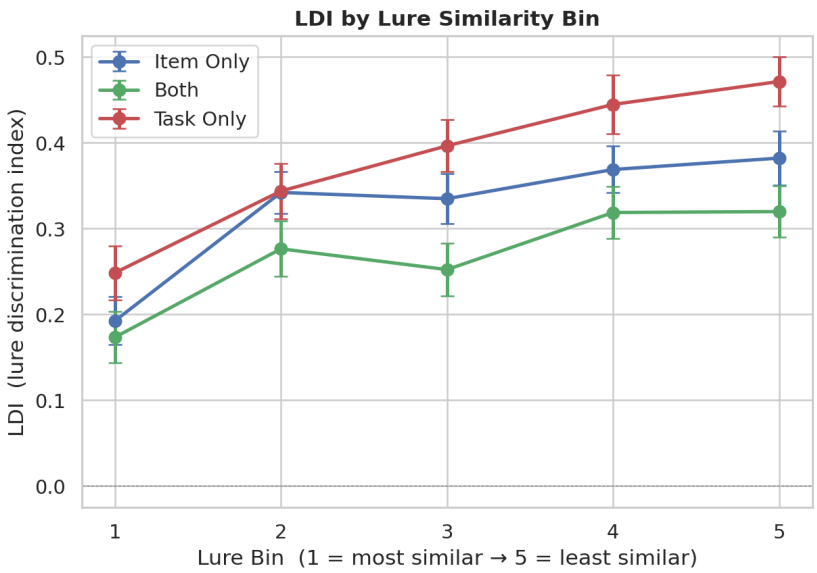


Figure 5: LDI as a function of lure similarity bin (1 = most similar → 5 = least similar). Error bars are ±1 SEM.

3.5 Response Proportions

The full response distribution (Figure 6) shows that participants used all three response options across all groups. Targets predominantly received "old" responses, lures received a mix of "similar" and "old," and foils were mostly classified as "new," confirming appropriate task performance.

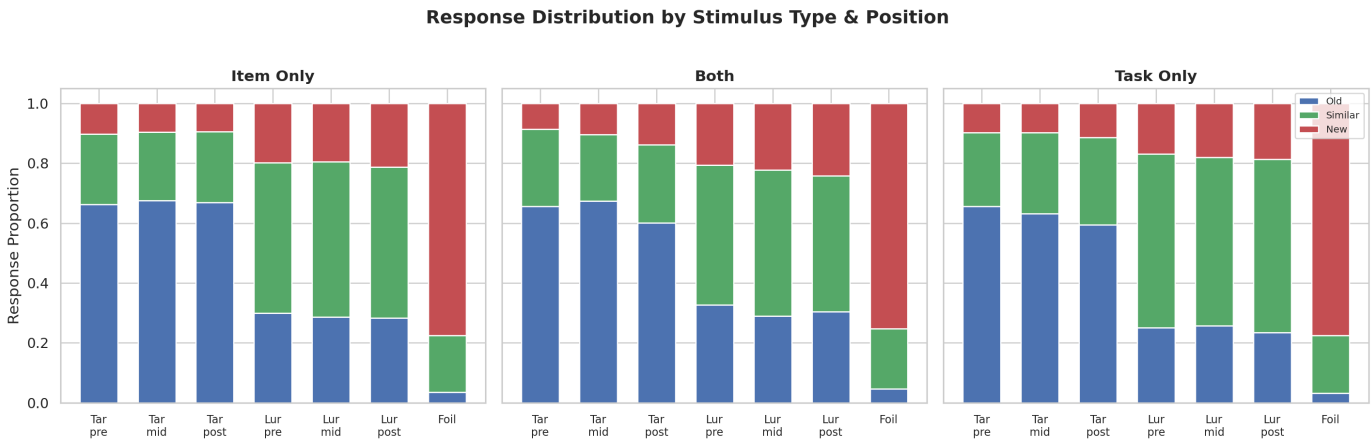


Figure 6: Response proportions (old/similar/new) by stimulus type and boundary position for each group.

3.6 Individual Variability and Encoding Accuracy

Considerable individual variability was observed in both REC and LDI (Figure 7). A floor effect was evident in a minority of participants, particularly in the Task Only group. Encoding accuracy was significantly higher in the Both ($M = 0.95$) and Task Only ($M = 0.94$) groups compared to Item Only ($M = 0.80$; both $p < .001$, $d > 3.8$), likely because "indoor or outdoor?"—the sole Item Only question—is harder to answer consistently for objects than the "shoebox" question.

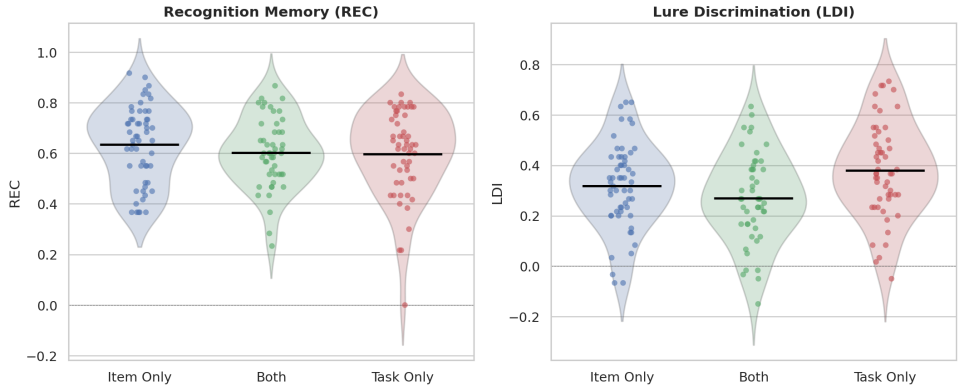
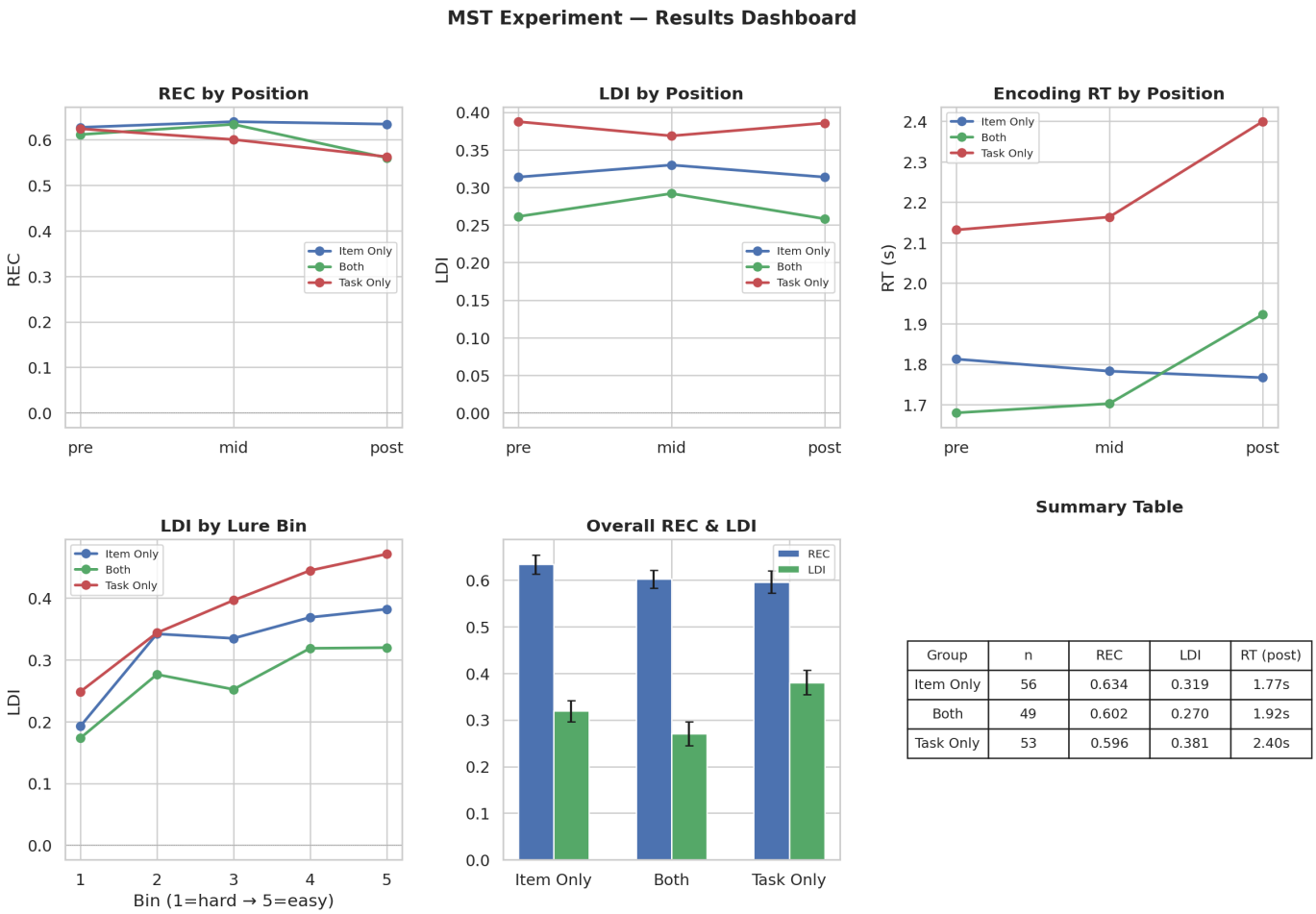


Figure 7: Individual participant REC and LDI scores by group (violin plot with strip overlay). Black horizontal lines indicate group means.

3.7 Summary Dashboard

Figure 8 provides a consolidated overview of all key metrics.



Group	n	REC	LDI	RT (post)
Item Only	56	0.634	0.319	1.77s
Both	49	0.602	0.270	1.92s
Task Only	53	0.596	0.381	2.40s

Figure 8: Summary dashboard of all key metrics across groups.

4 Conclusion and Future Directions

This replication partially supports the directional boundary effects reported by Morse et al. (2023):

1. **Post-boundary recognition impairment:** Successfully replicated. In the Both and Task Only groups, post-boundary items showed selectively reduced REC relative to both mid-event and

pre-boundary items, which did not differ from each other. This impairment was absent in the Item Only group, which lacked a task switch.

2. **Pre-boundary lure discrimination enhancement:** Not replicated. No significant position-based differences in LDI were observed in any group. This null result may reflect differences in paradigm (7 items per event vs. 10 in Morse et al. 2023, "shoebox" vs. "natural/manmade" question, or the addition of a "similar" response option at test).

3. **Post-boundary RT slowing:** Strongly replicated. Both the Both and Task Only groups showed robust post-boundary RT increases (all $d > 1.1$), consistent with the need to update event representations at boundaries. Item Only showed no effect.

Interpretation. The convergence of RT slowing and recognition impairment at the post-boundary position suggests that the cognitive cost of processing a new event context—likely involving an updating of the active event model (Zacks et al., 2007; Radvansky, 2012)—comes at the expense of encoding fidelity for the immediately following item. The absence of these effects in the Item Only group confirms that a task switch (not merely a perceptual shift) is necessary to produce these boundary effects, consistent with Experiments 2 and 3 of Morse et al. (2023).

Interestingly, *both* the Both and Task Only groups showed comparable REC impairment and RT slowing, diverging from Morse et al. (2023) where only the conjunction condition (Experiment 1) showed the full pattern. This may suggest that a task switch alone is sufficient to generate post-boundary recognition costs in the present paradigm.

Future directions.

- Apply multiple-comparison corrections (Bonferroni or FDR) to the pairwise tests.
- Conduct mixed-model ANOVAs with group (between) × position (within) factors.
- Explore whether LDI effects emerge when analysed by lure bin—the hardest bins (1–2) may show boundary effects that are diluted when averaging across bins.
- Investigate the relationship between encoding RT and subsequent recognition on a trial-by-trial basis.
- Consider exclusion criteria (e.g., removing participants at floor or ceiling) to reduce noise.

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Individual Contributions

- **Ananthula Sai Ravichandra:** Developed the data-processing pipeline (`mst_analysis_updated.py`), including CSV parsing, metric computation (REC, LDI, encoding RT), and group-level aggregation. Contributed equally to statistical analysis design and interpretation of results.
- **Krishiv Gupta:** Developed the visualization and statistical-testing pipeline (`mst_plots_updated.py`), including all 10 figures, within-group pairwise tests, between-group comparisons, and per-group boundary-effect computations. Contributed equally to statistical analysis design and interpretation of results.
- **Mahendraker Arya:** Led the report writing and literature review, drafting the Introduction, Methods, Results, and Conclusion sections. Contributed equally to statistical analysis design and interpretation of results.

All three members contributed equally to experiment design, data collection, and final review of the manuscript.